

Human-in-the-Loop Analytical Leadership

A professional method for directing
AI-assisted analytical work.
Demonstrated through a synthetic financing case.

Andrew R. Goad

- 01 **AI**
accelerated the work.
- 02 **Human judgment**
directed the work.
- 03 **Evidence**
governed the work.
- 04 **Separate challenge**
strengthened the work.
- 05 **Teaching**
made it transferable.

THE CASE STUDY

Merchant Sales-Based Financing (MSBF)

Explore the strategy simulator and published project evidence

github.com/andrew-goad/merchant-sales-based-financing-strategy-simulator

Agree on direction. Delegate the work. Retain judgment.

This instruction followed context gathering and strategy refinement, before execution.

ORIGINAL HUMAN INSTRUCTION / AUG. 28, 12:02 PM / EXCERPT

One more thing before we begin. You have ingested the entire GitHub (including project history/lineage), data, Power BI, Campaign work, Module 3 work, credit decisioning strategy simulator code. You have all the keys to the kingdom, so to speak to make this successful. I agree with the direction you've outlined. I want to do this work in as few chats prompt interactions as possible and in as few days as possible. Issues I've noticed in the past - We do not want to waste time/resources on a prompt that is too large and doesn't complete. We also do not want to spend time doing too little in each response and taking 100 chats to do something we could have done in 5. We want to maximize what we accomplish with each prompt. I authorize you to move ahead in leaps and only stop when you need a key decision or task from me. If it's something we've already agreed on in the strategy, do not raise that to the level of a decision. I agree with your recommendations now and the direction we've set now. We do not have to establish the full roadmap now, but please keep that as a governing principle and know that we will not go back to audit until just before republishing.

CONTEXT SUPPLIED

EXECUTION DELEGATED

JUDGMENT RETAINED

What evidence was there - and why was it blocked?

Merchant 003: Baseline provides a detailed write-up. Recession Energy names a block without explaining the evidence behind it.

ORIGINAL HUMAN PROMPT / EXCERPT

Here is Merchant 003. You see detailed narrative for Baseline, but the same required underwriting was blocked... for recession energy. I need more than that. What evidence was there, and why did it get blocked before a complete operating assessment could be produced. We need that why in our return. It's not sufficient to say, "Required source or verification evidence was insufficient to support a complete operating assessment." as you suggest.

ORIGINAL REPLY / SAME CAPTURE / EXCERPT

Yes. Merchant 000003 exposes the next issue: we are still narrating the status of the pipeline rather than the evidence chain that caused that status.

For 000003 under Recession Energy, we actually know considerably more than the page is saying. The selected row has high source confidence (98.6%), clear verification, fraud Tier 1, monitored processor continuity, complete feature-completeness status, \$28.6K average available balance, no negative-balance days, and no NSF's. What disappears under stress is the merchant's 30-day eligible-sales measure and annualized eligible sales. Once `avg_daily_eligible_sales_30d` is unavailable, M1.10 cannot estimate daily operating cash flow; the governed capacity engine explicitly sets capacity evidence to `BLOCKED` when that sales measure is null. Coverage and residual cash flow therefore cannot be calculated, and the missing capacity evidence cascades into resilience/risk/loss/economics being unusable.  Pasted text(20260827-165737)

AUG. 27 / 11:54 PM

THE HUMAN INTERVENTION

Baseline describes the merchant's operating facts. Recession Energy says evidence was blocked. The human question asks what was available, what was missing and why that stopped the assessment.

PRACTICE TO REUSE

A blocked assessment still needs a reason the reader can trace to source evidence.

Baseline explained the assessment. The stress view did not.

One synthetic applicant under two economic scenarios. These are alternatives, not successive events.

BASELINE / ORIGINAL WRITE-UP

① **Supporting detail is visible.**

Merchant operating evidence

The merchant showed a volatile operating pattern, averaging \$4.0K in eligible sales per day, with no recent NSF events and no negative-balance days. Average available balance was \$99.3K. Processor continuity required monitoring. The evidence was high-confidence and complete, with clear verification.

Structure assessment narrative

At the requested structure, the governed capacity assessment was unaffordable: modeled operating cash flow covered total daily obligations 0.73x, producing a \$360 per-day shortfall, with a post-financing liquidity buffer of \$88.5K. Integrated risk was moderate (Tier 2; score 21.1). Comparative expected loss was 6.37%. Risk-adjusted contribution was \$2,168 (21.5% annualized), above the governed hurdle.

P03 Decision Readout

The requested structure exceeded sustainable cash-flow capacity; no supportable alternative structure was authorized.

Sales, balances and cash-flow calculations give the reader evidence behind the assessment.

RECESSION ENERGY / ORIGINAL WRITE-UP

② **The explanation stops at “blocked.”**

Merchant operating evidence

Required underwriting evidence was blocked before a complete operating assessment could be produced.

Structure assessment narrative

Required underwriting evidence was blocked before the governed capacity, risk, comparative-loss, and economic assessment could be completed.

P03 Decision Readout

One or more governed eligibility or structure gates failed; no supportable structure was authorized.

Which evidence was unavailable? Why was it unavailable? Why did that stop the assessment?

A blocked assessment still needs an evidence-based explanation.

The stress view now explains the block.

Later Recession Energy result: the write-up identifies the missing sales evidence and why the assessment cannot proceed.

ORIGINAL POWER BI DETAIL / EDITORIAL CUES 1-2

DECLINED

Evidence decline

Merchant request

Applied	Requested	Use	Channel	Remittance	Payback multiple	Payoff days	Repayment
7/23/2026	\$61,300.00	Working capital	Digital direct	13.67%	1.15	60	\$70,644.20

Proposed structure

Amount	Remittance	Payback multiple	Horizon days	Repayment	Funding date
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Merchant operating evidence

1

This is a new lower-middle-market healthcare-services merchant sourced through the digital-direct channel. The merchant profile was available, but the governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days. Because that gap prevented a reliable capacity and risk assessment, the platform stopped rather than treating missing evidence as weak performance.

Risk & economics assessment

The merchant requested \$61,300 at a 13.67% remittance rate, 1.15x payback and a 60-day horizon, for \$70,644 in total repayment. A reliable structure assessment could not be completed because the governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days. This is an evidence-based stop, not a conclusion that every possible structure would fail.

Decision

2

Declined because required evidence was unavailable. The governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days, so the platform could not establish a sufficiently reliable view of the merchant's operating position. No structure or funding decision was inferred from incomplete data.

1 THE SPECIFIC GAP

Only 28 of the required 30 sales-history days are usable. Two days were lost to processor outages.

2 THE EXPLAINED STOP

The missing sales history blocks the dependent assessment. That explains the stop; it does not prove every offer would fail.

Design the handoff before the rerun.

Plan how the work will run and what evidence it will leave, before asking AI to execute it.

ORIGINAL HUMAN INSTRUCTION / ATTACHED CONTEXT



MSBF_CAMPAIGN_SCALE_TO_M1_M2_...
Document



MSBF_CAMPAIGN_SCALE_TO_M1_M2_...
File

Let's think about evidence we want to capture in this re-run and how we do the re-run. We had a separate chat called MSBF - Campaign Scale ... that has been working towards a smoother way to perform a full re-run and what evidence is important. Here is the latest from that chat (although notably, audit drove it sort of madly into a spiral around overaggressive bootstrapping/keys...versus getting the job done). Is anything usable from that pursuit? I need to find a better way than running 230 odd programs back to back to ... We will follow my no cold handoffs approach, but in this instance, I will be heavily involved in filling audit's shoes upfront and we'll pass everything we did to audit afterwards to approve before publishing. The most important thing is accuracy and speed and maintaining the high professionalism you see in my credit decisioning and msbf work to date. This should be exceptional.

AI RESPONSE / VERIFIED TRANSCRIPT

“The rerun should be one command, not 230 manual executions”

The proposed approach combines one controlled run with ordered execution and traceable evidence underneath.

PRACTICE TO REUSE

Carry forward useful controls while reducing avoidable execution burden.

Carry the reporting feedback back into the source strategy.

The reporting chat defines what users need to understand. The owner carries those requirements back to the build chat.

REPORTING RESPONSE / ORIGINAL REQUIREMENT EXCERPT

Power BI should ultimately be able to show:

Requested structure → why it failed or became marginal → feasible structure search → candidate alternatives → governing constraints → selected counteroffer → economics/risk improvement → review/authorization outcome

HUMAN RETURN HANDOFF / ORIGINAL PROMPT + ATTACHMENT

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Document

Feedback from our Power BI chat before we embark:

REMEDATION RESPONSE / ORIGINAL OPENING EXCERPT

This Power BI feedback is highly usable, and I would incorporate most of it before we write the successor strategy. It does not materially change the direction we were headed; it sharpens the evidence contract so that we do not repeat the same mistake—building correct-enough strategy logic but leaving downstream consumers unable to explain it.

An accepted build still has to make business sense.

A separate project review had accepted the build with observations. Later dashboard evidence raised a new monitoring question.

HUMAN FOLLOW-UP / INCLUDES QUOTED AI ANALYSIS

let's not leave this. let's dive in to understand it: One observation, not a defect

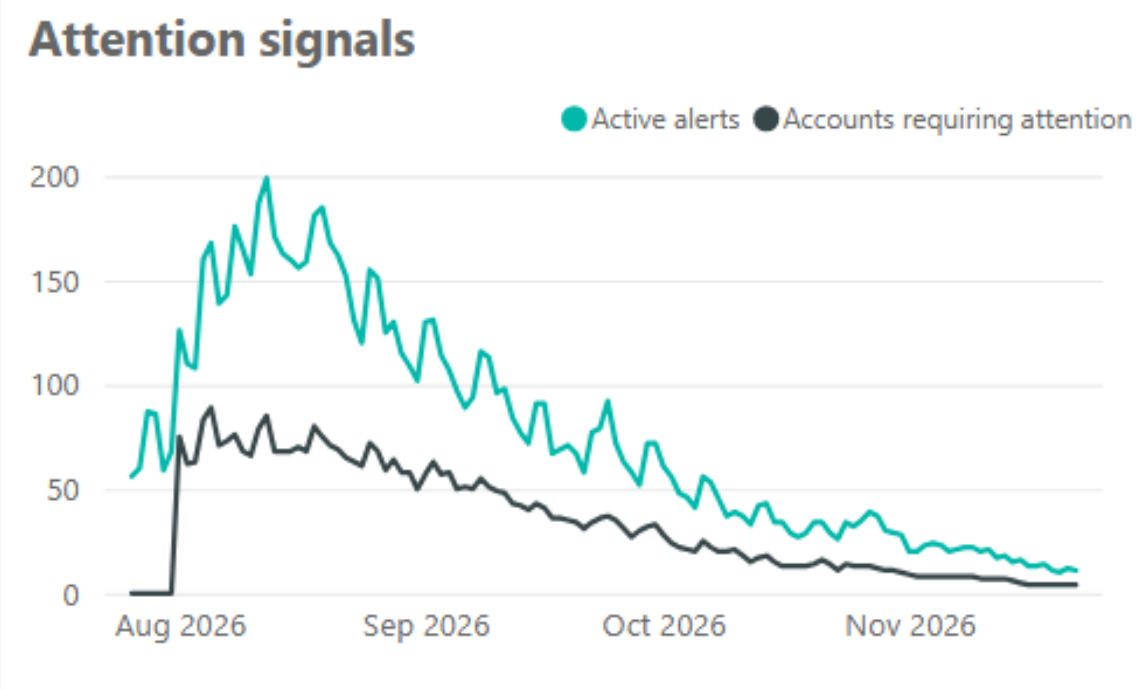
The P02 Attention signals chart has much larger absolute counts than the predecessor page. That initially looks dramatic, but the legend now explicitly says Active alerts and Accounts requiring attention, and the successor population is also much larger. The evidence shows those signals subsequently resolve toward zero. I would leave it exactly as built rather than normalize or suppress the behavior. first, specifically what are you comparing?

PRIOR AI RESPONSE / SAME CAPTURE

One observation, not a defect

The P02 Attention signals chart has much larger absolute counts than the predecessor page. That initially looks dramatic, but the legend now explicitly says Active alerts and Accounts requiring attention, and the successor population is also much larger. The evidence shows those signals subsequently resolve toward zero. I would leave it exactly as built rather than normalize or suppress the behavior.

THE PATTERN UNDER REVIEW / BEFORE



NEW QUESTION / AFTER ACCEPTANCE

Why do so many accounts appear to need attention? First, establish what the explanation is comparing.

Acceptance is a milestone. Keep judgment active as new behavior becomes visible.

Question whether the clock fits the funded account.

After diagnostic work and repeated checks, the human hypothesis asks whether newly funded accounts are being judged too early.

ORIGINAL HUMAN HYPOTHESIS

Tue, Sep 1 at 8:54 PM

is it possible that we are starting the 14 day timer too early after funding. perhaps the first month issue is related to funding timing and really a false positive. i do not want to delay this high priority issue. we need a policy/database solution now.

ORIGINAL AI SOURCE FINDING / SAME CAPTURE

So yes: the current pace-based deterioration clock becomes severe very early in a newly funded account's life.

More importantly, the current method compares two different concepts. Actual remittance is correctly sales-linked—`eligible POS sales × remittance rate`. `MSBF_R1_SUCCESSION_REPOSITORY_CA...` But cumulative expected remittance is a fixed straight-line benchmark using `implied_daily_collection_amount × monitoring_day_index`. `MSBF_R1_SUCCESSION_REPOSITORY_CA...` Then WATCH starts on day 7 and UNDERPERFORMING/SEVERE pace logic starts on day 14. `MSBF_R1_SUCCESSION_REPOSITORY_CA...`

That combination is the issue I would correct now—not funding dates themselves.

SEPT. 1 / 8:54 PM

THE CAUSAL QUESTION

Does the system allow enough time after funding, and is the repayment benchmark appropriate? The response identifies a broader problem in how pace is judged.

PRACTICE TO REUSE

Offer a mechanism the source logic and retained evidence can test.

Request a bounded challenge. Receive a recorded decision.

The second focused review, identified as R26, checks the monitoring correction, its outputs and the permitted next step.

ORIGINAL AUDIT REQUEST / EXCERPT

COPY/PASTE PROMPT — BOUNDED R26 MONITORING-CALIBRATION SUCCESSOR AUDIT

Perform a focused Independent Audit of:

MSBF_R26_FOCUSED_INDEPENDENT_AUDIT_SUBMISSION_20260902.zip

Before deciding:

1. verify the outer manifest/checksum/receipt;

2. ingest 00_START_HERE.md, all docs/, prior_authority/, evidence/, and required inputs/;

3. treat the prior R23 decision as controlling predecessor acceptance authority;

4. independently verify the physical R26 R2 failure result and final resume PASS result rather than reconstructing them from this prompt;

5. preserve accepted/public predecessor authority unless this focused audit explicitly accepts R26 as the next technical successor baseline.

Decision requested: accept/reject/hold R26 M2.5–M2.12 as a technical successor baseline, including the owner-approved monitoring policy, the retained-final resume method, corrected M2.11 semantic identity boundary, 52,152/12,000/1,040 simulation populations, and 70/70 G3-v2 certification.

Do not authorize GitHub publication, public supersession activation, or Power BI publication through this audit. If R26 passes, authorize only the next bounded integration step: regenerate the version-separated Power BI/repository candidate from exact R26 physical facts and submit that delta for appropriate review before publication.

ORIGINAL AUDIT DECISION / EXCERPTS

R26 focused successor audit completed

Decision ID: IA-D-R15-M12-001

Governing operating model: IA-GOV-20260822-001

Overall disposition: PASS_WITH_OBSERVATIONS

Acceptance includes observations. The review distinguishes the corrected policy, reproduced outputs, failed-run history and permission to integrate.

Governing decisions	
Matter	Independent disposition
Owner-approved R26 M2.5 monitoring policy	PASS
Physical R2 M2.5–M2.10 business outputs	PASS
R26 R2 runtime as a whole	FAIL — HISTORICAL PROVENANCE
Retained-final clone-and-resume architecture	PASS_WITH_OBSERVATIONS
R26 M2.11 eight-profile simulation	PASS
R26 M2.12 / G3-v2 certification	PASS
R26 technical successor baseline	PASS
Version-separated repository / Power BI integration	AUTHORIZED NEXT STEP

Verify the change. Reconcile what should stay the same.

Check the before-and-after monitoring results for the same 130 funded scenario paths. Confirm that cash history and end-of-period outcomes remain unchanged.

103 BASELINE FUNDED PATHS

41 → 20

BEFORE → AFTER

Baseline: peak adverse accounts

27 STRESS FUNDED PATHS

11 → 4

BEFORE → AFTER

Stress: peak adverse accounts

Adverse: underperforming, severe shortfall or dormant/no remittance; WATCH is excluded.

END-OF-PERIOD OUTCOME	VALUE - UNCHANGED BY CALIBRATION
Baseline paths paid off by period end	99 of 103 paths
Stress paths paid off by period end	27 of 27 paths
Baseline severe accounts at period end	4
Baseline attention exposure at period end	\$7,823.28

The policy changes when repayment pace is flagged; cash history and end-of-period results stay the same. This establishes better calibration; it does not establish improved collections.

Use the same questions on the next analytical problem.

The case becomes a working discipline that another team can inspect and apply.

Frame	What business behavior needs a better explanation?	Trace	Which source information and rule produced the result?
Challenge	Does the accepted rule make sense for the problem?	Test	Which hypothesis separates the competing explanations?
Reconcile	Do the full set of records and the facts that should stay unchanged still agree?	Review	What can a separate reviewer verify, and what next step is permitted?
Return	Does the report explain what the evidence actually supports?	Teach	Can another person explain the result and its limits?

Stay accountable across the loop, including after an acceptance milestone.

Do not stop at the output. Teach the reason behind it.

Ask what the system has established.

Check what remains unproven.

Keep judgment accountable to evidence.

Andrew R. Goad

Human-in-the-Loop Analytical Leadership

For leaders who need analytical systems they can challenge, explain and trust.

EXPLORE THE PROJECT AND PUBLISHED EVIDENCE

github.com/andrew-goad/merchant-sales-based-financing-strategy-simulator